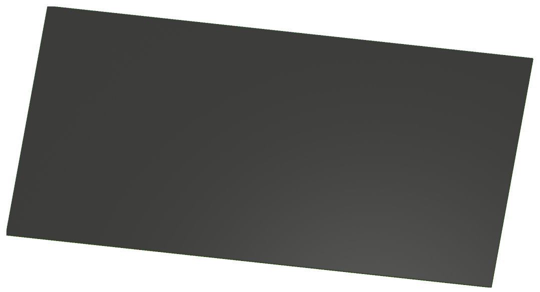
Mantle

Variant: DRAFT

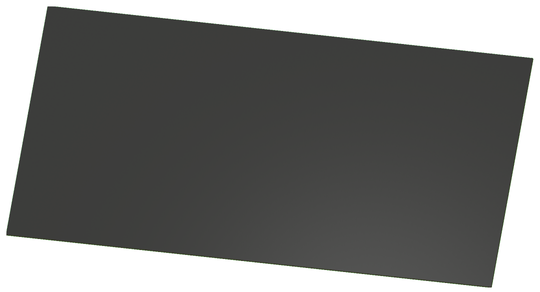
2026-02-03

Rev + (Unreleased)

TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 03-Feb-2026

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

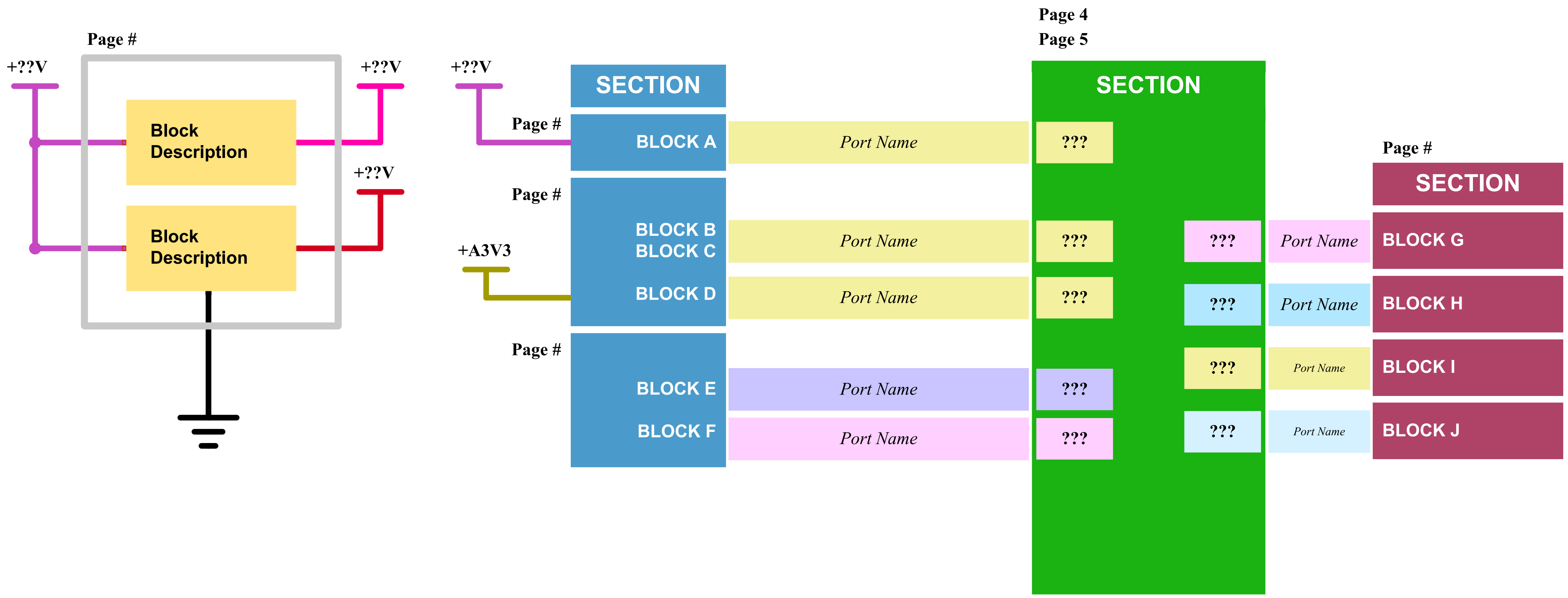
DESIGN NOTE: Example text for cautionary design notes.

DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

Comments:	Company: SICEMIC		Variant: DRAFT	Git Hash: a63fb32
	Board Name: **Mantle**		Project Name: **Arctic Seismic Sensor**	
Sheet Title:	File Name: seismic_sensor.kicad_sch	Designer: Arnór Aðalsteinsson	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /		Reviewer:	Size: **A3**	Sheet: **1** of **7**

[2] Block Diagram

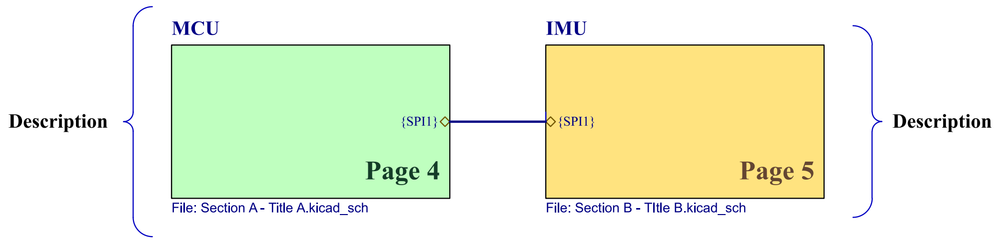


Target specifications:

Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

Comments:	Company: SICEMIC		Variant: DRAFT	Git Hash: a63fb32
	Board Name: **Mantle**		Project Name: **Arctic Seismic Sensor**	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Arnór Aðalsteinsson	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 7

[3] Project Architecture

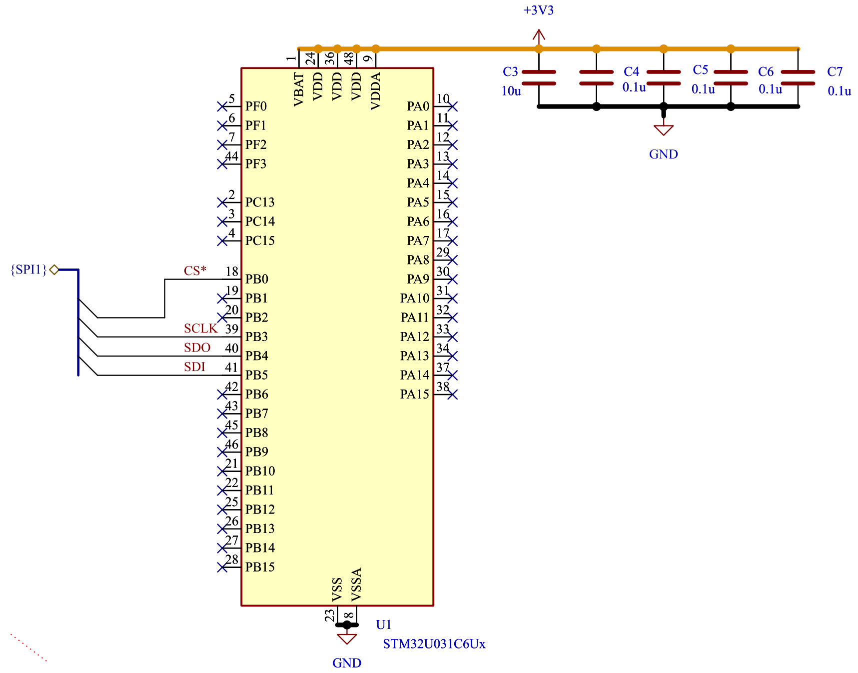


Comments:	Company: SICEMIC		Variant: DRAFT	Git Hash: a63fb32
	Board Name: **Mantle**		Project Name: **Arctic Seismic Sensor**	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Arnór Aðalsteinsson	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: **A3**	Sheet: **3** of **7**

[4] Sheet Title A

LAYOUT NOTE:
blablabla

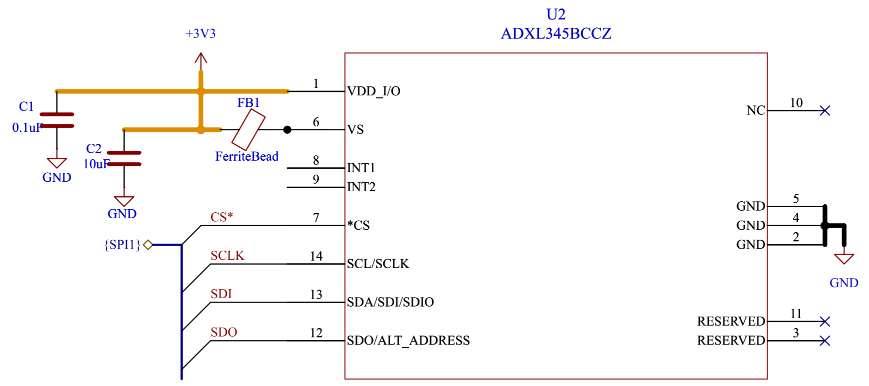
MCU Power Supply Scheme



MCU - GPIOs

Comments:	Company:		Variant:	Git Hash:
AN5346 STM32G474 Datasheet p.81 J. Pieper ADC investigation	SICEMIC		DRAFT	a63fb32
	Board Name: **Mantle**		Project Name: **Arctic Seismic Sensor**	
Sheet Title: Sheet Title A	File Name: Section A - Title A.kicad_sch	Designer: Arnór Aðalsteinsson	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/MCU/		Reviewer:	Size: A4	Sheet: 4 of 7

[5] Sheet Title B



Accelerometer

DESIGN NOTE:
blabla

DESIGN NOTE:
blabla

LAYOUT NOTE:
blabla

Block Description

Block Description

Block Description

Block Description

Block Description

Block Description

Block Description

Block Description

Block Description

Comments:

References:

Flexible I/O worked examples
Flexible I/O source configuration

Company: SICEMIC

Variant: DRAFT

Git Hash: a63fb32

Board Name: **Mantle**

Project Name: **Arctic Seismic Sensor**

Sheet Title: Sheet Title B

File Name: Section B - TItle B.kicad_sch

Designer: Arnór Aðalsteinsson

Date: 2025-01-12

Revision: + (Unreleased)

Sheet Path: /Project Architecture/IMU/

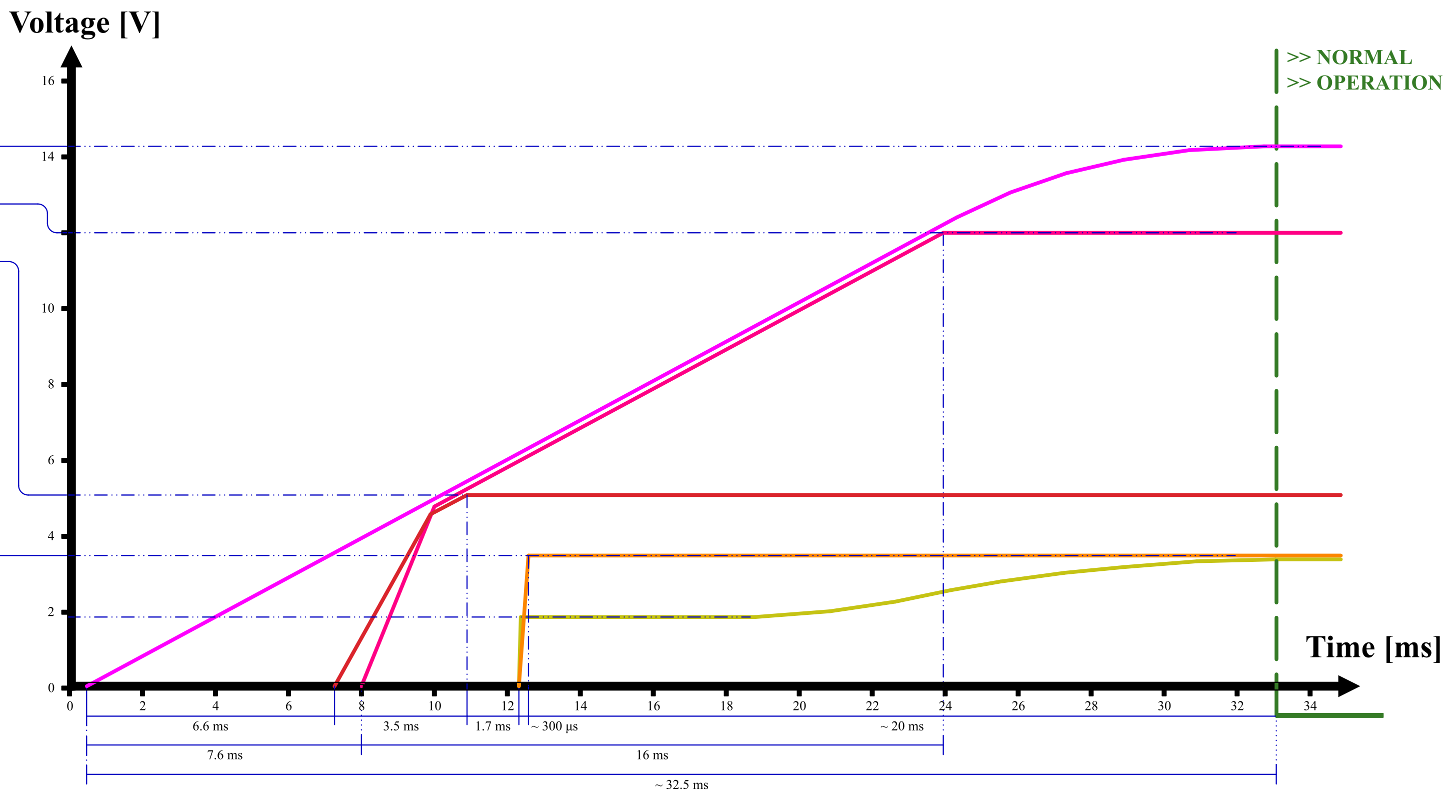
Reviewer:

Size: **A3**

Sheet: **5** of **7**

[6] Power - Sequencing

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%



Comments:	Company: SICEMIC		Variant: DRAFT	Git Hash: a63fb32
	Board Name: Mantle		Project Name: Arctic Seismic Sensor	
Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Arnór Aðalsteinsson	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 6 of 7

[7] Revision History

Comments:	Company: SICEMIC		Variant: DRAFT	Git Hash: a63fb32
	Board Name: **Mantle**		Project Name: **Arctic Seismic Sensor**	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Arnór Aðalsteinsson	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Revision History/		Reviewer:	Size: **A4**	Sheet: **7** of **7**